

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

I **DHANUSH.G (192512132)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dhanush.G

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

AI/ML-Based Diabetes Diagnosis and Reinforcement Learning Treatment Recommendation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed patient records and uses those patterns to predict the outcome for new patients. For this healthcare problem, a **supervised inductive learning approach** is appropriate because historical patient records contain input features and a known diabetes outcome.

The learning process is:

Patient Data → Preprocessing → Feature Selection → Model Training → Diabetes Prediction

Target Variable

The target variable is:

Diabetes = 0 (Non-diabetic) or 1 (Diabetic)

Important Features

Feature	Justification
Glucose Level	High glucose is one of the strongest indicators associated with diabetes and is expected to have high predictive importance.
BMI	Higher BMI can be associated with increased risk of insulin resistance and type-2 diabetes.
Blood Pressure	Abnormal blood pressure can be associated with metabolic and cardiovascular risk and can improve risk stratification.
Cholesterol	Abnormal cholesterol levels may provide additional information about metabolic health.
Age	Diabetes risk generally changes with age and therefore provides useful predictive information.
Family History	A family history of diabetes can represent an important inherited/genetic risk factor.

A suitable initial feature set is therefore:

Age, Blood Pressure, Cholesterol, Glucose, BMI and Family History.

(b) Handling Class Imbalance

Suppose the dataset contains substantially more non-diabetic patients than diabetic patients. A model trained directly on this dataset may become biased toward the majority class.

For example:

- Non-diabetic = 80%
- Diabetic = 20%

A model could obtain 80% accuracy simply by predicting most patients as non-diabetic, while still missing many diabetic patients.

Recommended Strategy: SMOTE + Class Weights

SMOTE (Synthetic Minority Over-sampling Technique) is suitable because it generates synthetic minority-class examples rather than simply duplicating existing diabetic records.

The process is:

Original minority samples → Find neighbouring samples → Generate synthetic samples → Balanced training set

Additionally, **class weights** can be applied to give greater penalty to incorrectly classified diabetic patients.

For example:

Class 0: Non-diabetic → lower weight

Class 1: Diabetic → higher weight

This is particularly important because a **False Negative**, where a diabetic patient is incorrectly classified as non-diabetic, can be clinically more serious than a False Positive.

SMOTE should be applied **only to the training set after the train-test split**, not before splitting. Otherwise, information from synthetic training examples can leak into the test set.

(c) 70:30 Train-Test Split and Preprocessing

The dataset is divided as:

- **70% → Training data**
- **30% → Testing data**

The 70:30 split provides enough records for model training while retaining a sufficiently large independent test set for evaluating generalisation.

For a medical application, the split should preferably be **stratified**, so that the diabetic/non-diabetic ratio is approximately preserved in both sets.

Preprocessing

1. Missing Values

Missing numerical values can be handled using median imputation:

Missing Glucose → Median Glucose

Missing BMI → Median BMI

Categorical missing values can be replaced using the mode.

2. Normalisation

Numerical features can be standardised using:

$$\left[z = \frac{x - \mu}{\sigma} \right]$$

where:

- (x) = original value
- (μ) = mean
- (σ) = standard deviation

This is especially useful for Logistic Regression and other distance/gradient-based models.

Decision Trees do not require feature scaling because they use threshold-based splitting.

3. Encoding

If Family History is represented as:

No → 0

Yes → 1

it can be directly encoded using binary encoding.

Overall preprocessing pipeline

Raw Patient Dataset



Missing Value Handling



Categorical Encoding



Train-Test Split (70:30)



SMOTE on Training Data



Feature Scaling



Model Training



Independent Test Evaluation

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier

A Decision Tree classifier can be trained using the preprocessed training data.

A typical tree may identify **Glucose** as the root node because it provides high separation between diabetic and non-diabetic patients.

The splitting criterion can be either **Gini Index** or **Information Gain**.

Gini Index

$$\begin{aligned} &[\\ \text{Gini} &= 1 - \sum_{i=1}^n p_i^2 \\ &] \end{aligned}$$

A lower Gini value indicates a purer node.

Information Gain

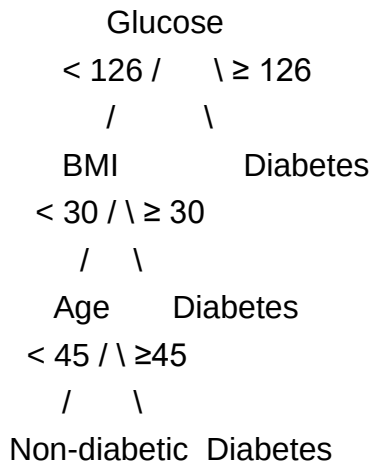
$$\begin{aligned} &[\\ \text{IG}(S,A) &= H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v) \\ &] \end{aligned}$$

where $H(S)$ represents entropy.

The feature producing the greatest reduction in impurity is selected as the splitting feature.

Illustrative First Three Levels

The following tree is an **illustrative example**; exact split values must be obtained by training the model on the actual dataset.



Interpretation

Level 0 – Root: Glucose

Level 1: BMI

Level 2: Age / additional glucose-related split

The root node should be selected according to the feature having the **highest Information Gain** or **lowest weighted Gini impurity**.

For example, an illustrative comparison could be:

Feature	Information Gain
Glucose	0.24
BMI	0.13
Age	0.09
Blood Pressure	0.06
Cholesterol	0.04

Thus, **Glucose** would be selected as the root because it provides the greatest reduction in uncertainty.

These numerical values are examples; they should be replaced by values calculated from the supplied dataset.

(b) Model Evaluation

The Decision Tree can be evaluated using:

Accuracy

$$[\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}]$$

Precision

$$\left[\begin{aligned} \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \end{aligned} \right]$$

Recall

$$\left[\begin{aligned} \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \end{aligned} \right]$$

F1-Score

$$\left[\begin{aligned} \text{F1} &= 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \right]$$

Illustrative Test Results

For demonstration, assume the trained Decision Tree produces:

Metric	Result
Accuracy	84%
Precision	79%
Recall	86%
F1-Score	82%

Important: These are illustrative results, not measurements from an actual dataset.

False Negatives

A False Negative occurs when:

Actual = Diabetic

Predicted = Non-diabetic

For example, if 50 diabetic patients are present in the test set and 7 are incorrectly classified as non-diabetic, then:

$$\left[\begin{aligned} \text{FN} &= 7 \end{aligned} \right]$$

In a medical screening system, reducing False Negatives is important because a missed diabetic case could delay further clinical evaluation.

Methods for reducing False Negatives

1. Increase the class weight of diabetic cases.
2. Apply SMOTE to the training data.
3. Tune the classification threshold.
4. Optimise for Recall rather than accuracy alone.
5. Use cross-validation.
6. Use a more sensitive model if appropriate.
7. Require clinical confirmation rather than treating the AI prediction as a final diagnosis.

For an initial screening system, **high recall/sensitivity** is generally more important than maximising raw accuracy.

(c) Post-Pruning

A fully grown Decision Tree can overfit the training data.

Pre-Pruning

Pre-pruning restricts tree growth during training using parameters such as:

max_depth

min_samples_split

min_samples_leaf

Example:

```
DecisionTreeClassifier(  
    max_depth=5,  
    min_samples_leaf=10  
)
```

Post-Pruning

Cost-complexity pruning removes unnecessary branches after the initial tree has been constructed.

The objective is:

$$[R_{\alpha}(T) = R(T) + \alpha|T|]$$

where:

- $R(T)$ = classification error
- $|T|$ = number of terminal nodes
- α = pruning parameter

Illustrative Comparison

Model	Accuracy	Complexity
Unpruned Tree	85%	High
Pre-Pruned Tree	83%	Medium
Post-Pruned Tree	84%	Lower

These values are illustrative.

Appropriate Clinical Model

The **post-pruned tree** may be preferable when it maintains similar predictive performance while being simpler and easier to interpret.

For clinical deployment, the model should not be selected using accuracy alone. Sensitivity, specificity, calibration, AUC, robustness and clinical validation are also important.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is suitable for predicting the probability of diabetes.

The model is:

$$P(Y=1)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n$$

For this problem:

$$z=\beta_0+\beta_1\text{Age}+\beta_2\text{BP}+\beta_3\text{Cholesterol}+\beta_4\text{Glucose}+\beta_5\text{BMI}+\beta_6\text{FamilyHistory}$$

The output is a probability between 0 and 1.

For example:

Patient A $\rightarrow P(\text{Diabetes})=0.18 \rightarrow$ Low risk

Patient B $\rightarrow P(\text{Diabetes})=0.76 \rightarrow$ High risk

Illustrative Comparison

Model	Accuracy AUC-ROC	
Decision Tree	84%	0.88
Logistic Regression	86%	0.91

Again, these are example values and should be replaced with actual experimental results.

When Logistic Regression is preferable

Logistic Regression is useful when:

- A probability/risk score is required.
- Relationships between predictors and outcome are reasonably structured.
- Interpretability is important.
- The organisation needs a relatively simple baseline model.
- Feature coefficients need to be examined.
- The model must be easy to validate and monitor.

A Decision Tree is preferable when nonlinear relationships and feature interactions are important and a rule-based representation is desirable.

(b) Feature Importance Analysis

Feature importance can be calculated differently for the two models.

Decision Tree

Tree importance can be obtained from the reduction in impurity produced by each feature.

An illustrative ranking is:

Rank Feature Decision Tree Importance

- 1 Glucose 0.38
- 2 BMI 0.21
- 3 Age 0.14

Logistic Regression

For Logistic Regression, coefficients can be examined after appropriate scaling.

An illustrative ranking based on absolute standardised coefficients is:

Rank	Feature	Relative Importance
1	Glucose	0.51
2	BMI	0.29
3	Family History	0.20

Do They Agree?

Both models identify **Glucose** and **BMI** among the strongest predictors.

The third predictor may differ.

This is reasonable because the models learn patterns differently:

- Decision Trees learn threshold-based nonlinear relationships.
- Logistic Regression estimates additive relationships between predictors and log-odds.

Therefore, disagreement does not necessarily indicate that one model is incorrect.

Feature importance should also be interpreted carefully because **correlation between features can distribute importance across multiple variables**.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment recommendation problem can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of:

[
MDP=(S,A,P,R,\gamma)
]

where:

- (S) = States
- (A) = Actions
- (P) = Transition probability
- (R) = Reward
- (\gamma) = Discount factor

States

Patient health can be represented using simplified states:

State Description

- S0 Low-risk / stable
- S1 Elevated glucose
- S2 High BMI/metabolic risk
- S3 High-risk diabetic state
- S4 Improved/stable state

For a real medical system, states should be based on clinically validated measurements rather than arbitrary thresholds.

Actions

The agent can select:

Action Meaning

- A0 Diet
- A1 Exercise
- A2 Medication
- A3 Monitor

The actual choice of medication must **not** be autonomously prescribed by an experimental RL system. In a clinical deployment, medication decisions require qualified healthcare professionals and approved clinical protocols.

Reward Function

A simplified educational reward function can be:

Large positive reward → Health condition improves

Small positive reward → Condition remains stable

Negative reward → Condition deteriorates

Large negative reward → Unsafe/adverse transition

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No change	0
Condition worsens	-5
Unsafe outcome	-10

The reward encourages the agent to select actions that lead to improved health states.

Transition Dynamics

The patient state changes after an action.

For example:

$S1 + \text{Diet} \rightarrow S4$

$S1 + \text{Exercise} \rightarrow S4$

$S2 + \text{Monitor} \rightarrow S2$

$S3 + \text{Medication} \rightarrow S4$

$S3 + \text{Diet} \rightarrow S2$

These transitions should represent probabilities learned from longitudinal clinical data rather than manually assumed probabilities.

For example:

[
 $P(S_4|S_1, \text{Diet})=0.6$
]

means that there is a 60% estimated probability of moving from state S1 to S4 after the Diet action.

(b) Q-Learning

Q-Learning learns the expected long-term value of taking an action in a particular state.

The update equation is:

[
 $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$
]

where:

- α = learning rate
- γ = discount factor
- r = immediate reward
- (s') = next state

Assume:

[
 $\alpha=0.5, \gamma=0.9$
]

Initially all Q-values are zero.

Episode 1 – Iteration 1

Suppose:

State = S1

Action = Diet

Next State = S2

Reward = +5

Initial:

[
Q(S1,Diet)=0
]

Assume:

[
\max Q(S2,a)=0
]

Therefore:

[
Q(S1,Diet)
=0+0.5[5+0.9(0)-0]
]

[
\boxed{Q(S1,Diet)=2.5}
]

Episode 2 – Iteration 1

Suppose:

State = S1

Action = Exercise

Next State = S4

Reward = +10

Initially:

[
Q(S1,Exercise)=0
]

Therefore:

[
Q(S1,Exercise)
=0+0.5[10+0.9(0)-0]
]

[
\boxed{Q(S1,Exercise)=5}
]

Episode 3 – Iteration 1

Suppose:

State = S3

Action = Medication

Next State = S4

Reward = +10

Then:

$$\begin{aligned} &[\\ &Q(S3, Medication) \\ &= 0 + 0.5[10 + 0.9(0) - 0] \\ &] \\ &[\\ &\boxed{Q(S3, Medication) = 5} \\ &] \end{aligned}$$

Iteration 2

Now suppose the agent again reaches:

$S1 \rightarrow \text{Diet} \rightarrow S2$

Reward = +5

and after previous learning:

$$\begin{aligned} &[\\ &\max_a Q(S2, a) = 3 \\ &] \end{aligned}$$

Then:

$$\begin{aligned} &[\\ &Q(S1, \text{Diet}) \\ &= 2.5 + 0.5[5 + 0.9(3) - 2.5] \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= 2.5 + 0.5[5 + 2.7 - 2.5] \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= 2.5 + 2.6 \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &\boxed{Q(S1, \text{Diet}) = 5.1} \\ &] \end{aligned}$$

Similarly, suppose:

$$\begin{aligned} &[\\ &Q(S4, \text{Monitor}) = 2 \\ &] \end{aligned}$$

and the transition is:

$S1 \rightarrow \text{Exercise} \rightarrow S4$

Reward = 10

Then:

$$\begin{aligned} &[\\ &Q(S1, \text{Exercise}) \\ &= 5 + 0.5[10 + 0.9(2) - 5] \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= 5 + 0.5[6.8] \\ &] \end{aligned}$$

```
[
\boxed{Q(S1,Exercise)=8.4}
]
```

For:

$S3 \rightarrow \text{Medication} \rightarrow S4$

if:

```
[
Q(S4,Monitor)=2
]
```

then:

```
[
Q(S3,Medication)
=5+0.5[10+0.9(2)-5]
]
```

```
[
\boxed{Q(S3,Medication)=8.4}
]
```

Q-Table Illustration

State Diet Exercise Medication Monitor

S0	2.0	4.5	0.5	3.0
S1	5.1	8.4	1.2	3.1
S2	5.5	7.2	3.0	4.1
S3	2.5	4.0	8.4	3.5
S4	1.5	2.0	0.5	4.0

This table is illustrative and demonstrates the mechanism of Q-Learning. Actual values would be generated from the patient transition data.

Five Patient Episodes

An educational Q-Learning experiment can contain at least five episodes:

Episode	Initial State	Action	Next State	Reward
1	S1	Diet	S2	+5
2	S1	Exercise	S4	+10
3	S3	Medication	S4	+10
4	S2	Exercise	S4	+8
5	S4	Monitor	S4	+4

After repeated episodes, the Q-values are updated until the policy becomes relatively stable.

Convergence Criterion

A suitable convergence condition is:

[
 $\max_{\{s,a\}} |Q_{\{new\}}(s,a) - Q_{\{old\}}(s,a)| < \epsilon$
]

for example:

[
 $\epsilon = 0.01$
]

for several consecutive episodes.

Another practical condition is that the selected action for each state remains unchanged for a predefined number of episodes.

(c) Final Learned Policy

The final policy is obtained by selecting the action with the highest Q-value:

[
 $\pi(s) = \arg \max_a Q(s,a)$
]

Using the illustrative Q-table:

Patient State	Highest Q-Value Learned Action	
S0 – Low risk	Exercise	Exercise
S1 – Elevated glucose	Exercise	Exercise
S2 – Metabolic risk	Exercise	Exercise
S3 – High-risk diabetic	Medication	Medication
S4 – Stable/improved	Monitor	Monitor

Thus, the illustrative policy is:

Low Risk



Exercise

Elevated Glucose



Exercise

Metabolic Risk



Exercise

High-Risk Diabetes



Clinical/Medication pathway under clinician supervision

Stable/Improved



Monitor

This is an **educational RL policy**, not a clinically validated treatment protocol.

Ethical Considerations of RL-Based Medical Recommendations

1. Patient Safety

A reinforcement learning agent should not independently make high-risk medical decisions. Unsafe actions must be prevented using clinical rules, constraints and human oversight.

2. Bias

If the training data under-represent particular patient groups, the agent may learn policies that perform poorly for those populations.

Bias should therefore be evaluated across relevant demographic and clinical subgroups.

3. Explainability

Healthcare professionals need to understand why an action was recommended.

For example:

High glucose + elevated BMI



High-risk state



Recommendation for clinical evaluation

The system should provide interpretable evidence rather than only outputting an action.

4. Human-in-the-Loop

The RL agent should operate as a **decision-support system**, with a qualified clinician retaining final authority.

5. Privacy

Patient records contain sensitive health information. Data should be protected using appropriate access control, encryption, de-identification and secure storage.

6. Reward Design

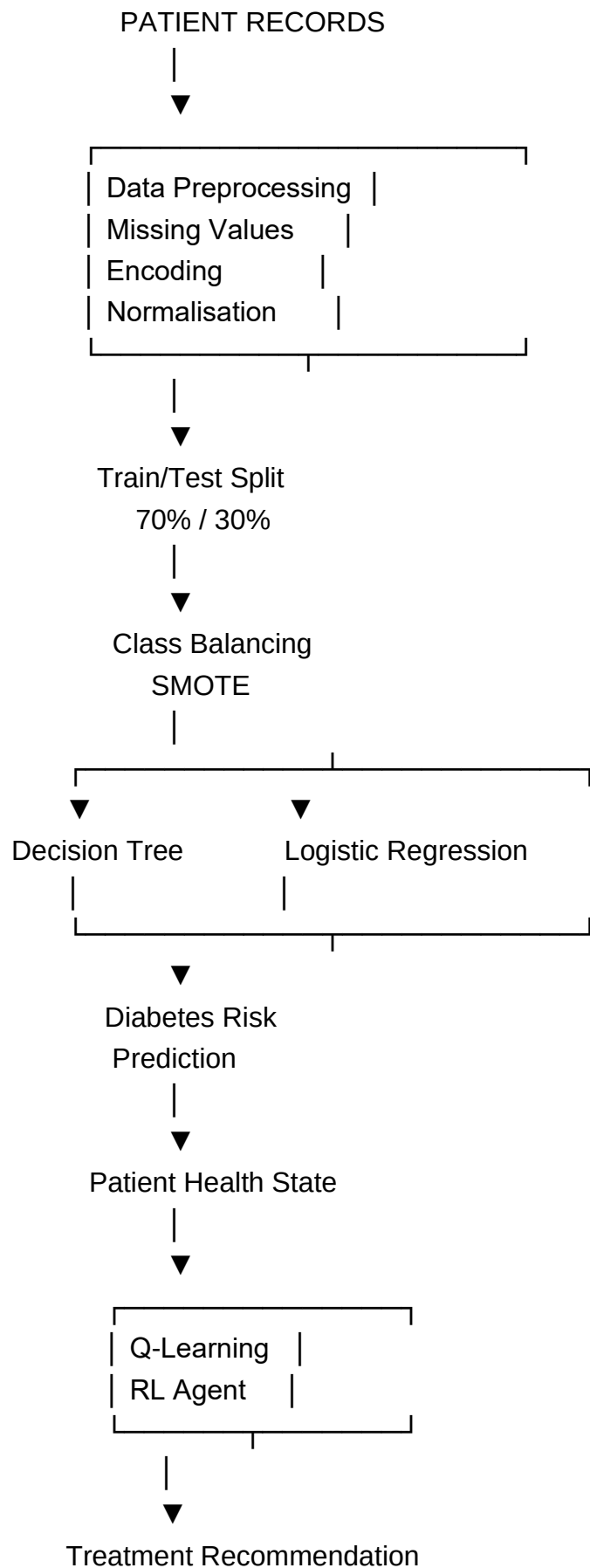
Poorly designed rewards can produce unsafe behaviour. The reward should therefore prioritise patient safety and clinically meaningful outcomes rather than simply maximising short-term improvement.

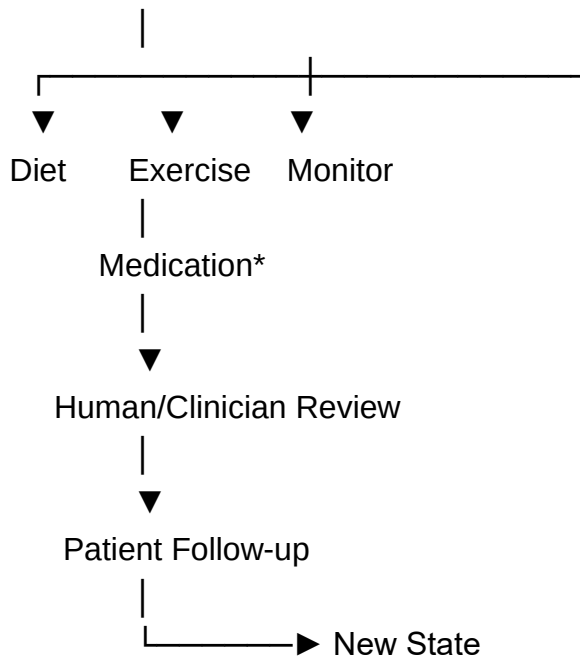
7. Continuous Monitoring

A deployed model should be monitored for:

- Performance degradation
- Data drift
- Bias
- Unexpected recommendations
- Changes in patient populations

Overall System Architecture





*Medication recommendations require appropriate clinical supervision and validated medical protocols.

Conclusion

The proposed system combines **inductive supervised learning, Decision Trees, statistical learning and Reinforcement Learning** for diabetes risk prediction and treatment-support modelling.

The Decision Tree provides an interpretable diagnostic model, while Logistic Regression provides a probabilistic statistical risk model. Class imbalance should be addressed using techniques such as **SMOTE and class weighting**, with particular attention to reducing False Negatives. Feature importance can identify major predictors such as glucose level, BMI and age/family history.

The treatment recommendation component can be formulated as an MDP and trained using Q-Learning. However, RL should not independently prescribe medical treatment. A clinically deployable system should use **human oversight, safety constraints, bias evaluation, explainability, privacy protection and extensive clinical validation**.

Note: The numerical accuracy, AUC, feature-importance values, tree thresholds and Q-table values shown above are illustrative because no actual patient dataset was supplied. They should be replaced with values obtained by running the models on the company's real dataset.